SRM Institute of Science and Technology

College of Engineering and Technology

B.Tech – CSE/ECE/BIOTECH

Academic Year: 2025-26 ODD **Semester:** VII **Mark:** 50 **Duration:** 100 min.

Subject Code: 21MEO101T **Title:** Fundamentals of Composite Materials

Prepared by: D. Selwyn Jebadurai **Type of Test:** FT II

Q. No.	Part A – MCQ (11 * 1 = 11 Marks)	M	BL	CO	PO
1.	Short fibre composites generally exhibit properties that are a) Nearly isotropic b) Perfectly anisotropic c) Extremely brittle d) More ductile than metals	1	2	1	1
2.	Which matrix type is most suitable for high-temperature applications? a) Polymer matrices b) Ceramic matrices c) Metal matrices d) Organic polymer blends	1	1	1	1
3.	The interface in composites is critical because a) It ensures chemical resistance b) It bonds the matrix and reinforcement, enabling load transfer c) It reduces the volume fraction of the reinforcement d) It lowers thermal conductivity	1	1	1	1
4.	In composite architecture, the role of reinforcement is to a) Act as a protective coating b) Provide strength and stiffness c) Reduce cost of manufacturing d) Increase ductility	1	1	1	1
5.	What is the composite architecture where layers of different materials are stacked and bonded together? a) Particulate composite b) Laminate composite c) Fiber-reinforced composite d) Whiskers composite	1	1	1	1
6.	Which type of reinforcement provides the best load transfer efficiency? a) Particles b) Flakes c) Continuous fibres d) Whiskers	1	1	1	1
7.	Which assumption is made in the rule of mixtures? a) Both fibre and matrix deform equally in transverse direction b) Perfect bonding exists at the fibre-matrix interface c) Fibers are discontinuous and randomly oriented d) Matrix is always stiffer than fibres	1	1	2	2
8.	Classical lamination theory (CLT) is mainly used to compute a) Heat transfer in composites b) Stress–strain response of multilayer laminates	1	1	2	2

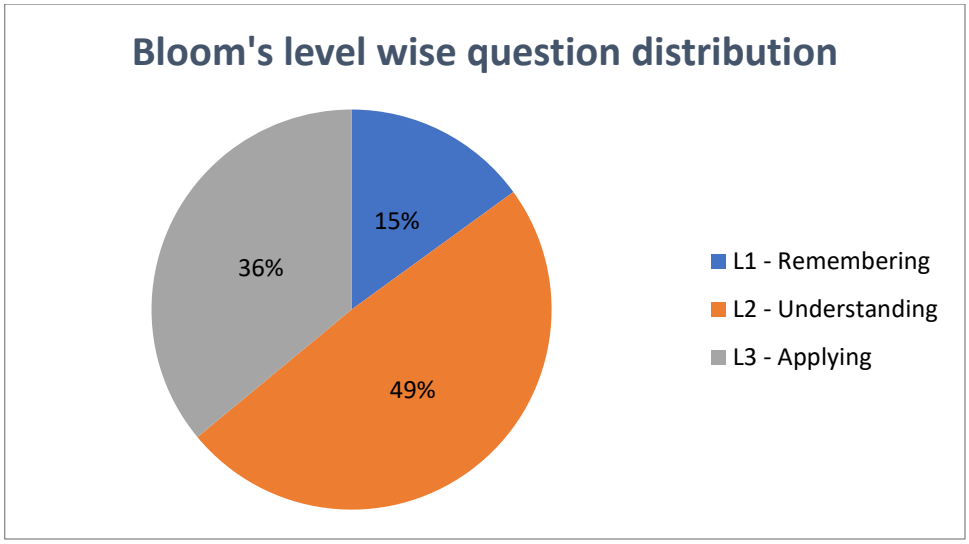
	c) Elastic constants of pure fibres d) Damage mechanisms in composites				
9.	Fiber pull-out during failure in composites is primarily governed by a) Fiber modulus b) Thermal conductivity of fibres c) Fiber-matrix interfacial strength d) Volume fraction of voids	1	1	2	2
10.	In a unidirectional fibre-reinforced composite, the transverse modulus is typically a) Greater than the longitudinal modulus b) Equal to the longitudinal modulus c) Less than the longitudinal modulus d) Independent of the constituent properties	1	1	2	2
11.	A lamina is considered orthotropic. How many independent elastic constants are needed to describe its in-plane behaviour? a) 2 b) 4 c) 5 d) 9	1	1	2	2
	Part B – 8 Mark (3 * 8 = 24 Marks) **Answer Any THREE questions**				
12.	Explain the concept of interfaces in composites. Discuss the different bonding mechanisms at the interface.	8	2	1	1
13.	Compare composite materials with conventional engineering materials in terms of properties and performance. Discuss the need for composites in modern engineering applications.	8	3	1	1
14.	Discuss the main damage mechanisms in fibre-reinforced composites under mechanical loading and explain how they affect structural performance.	8	3	2	2
15	Explain the steps involved in concept of classical lamination theory (CLT). What are the key assumptions of this theory?	8	2	2	2
	Part C (1 * 15 = 15 Marks)				
16 a)	(i) Derive the expression for longitudinal and transverse Young's modulus of an aligned fibre composite using the rule of mixtures. (ii) A unidirectional carbon/epoxy composite is made with 60% carbon fibre by volume. If the Young's modulus of the fibre is 350 GPa and that of the epoxy matrix is 3.5 GPa, Calculate the longitudinal modulus (E_1) and the transverse modulus (E_2) of the composite.	10 5	2 3	2 2	2 2
	(OR)				
16 b)	(i) Discuss the classification of composite materials based on their constituent phases, focusing on the roles and types of the matrix and reinforcement. (ii) Describe the primary characteristics, and the typical real-world applications of each type of composites.	10 5	2 3	1 1	1 1

Outcome Alignment Matrix:

QUESTION NUMBER	CO distribution				
	CO1	CO2	CO3	CO4	CO5
1	1				
2	1				
3	1				
4	1				
5	1				
6	1				
7		1			
8		1			
9		1			
10		1			
11		1			
12	8				
13	8				
14		8			
15		8			
16a	15				
16b		15			
Total	37	36			
%	51	49			

"**Quality Matrix**":

Question No.	**BL Distribution**			
	L1	**L2**	**L3**	**L4**
1	1			
2	1			
3	1			
4	1			
5	1			
6	1			
7	1			
8	1			
9	1			
10	1			
11	1			
12		8		
13			8	
14			8	
15		8		
16a		10	5	
16b		10	5	
Total	11	36	26	
%	15	49	36	



Prepared by:

(Kindly add name and signature)

Course Coordinator

(Kindly add name and signature)

Verified and approved by:

(Kindly add name and signature)

Signature of Professor In-charge

M – Marks for each question

CO – Course Outcome

BL – Bloom's Level

PO – Program Outcome

PI – Performance Indicator